

Geometry
Unit 1
Lesson 8 Practice

Name _____
Date _____
Period _____

Line segment KN is shown where $KR = 16x - 3$, $RW = WN$, $WN = 9x - 3$, $BR = \frac{1}{2}RW$, and $KN = 161$.



1. Find the value of x .
2. Find the length of $\overline{RW}$.
3. Find the length of $\overline{KW}$.
4. Find the length of $\overline{BR}$.
5. Use a compass and straightedge to make a copy of $\overline{HT}$.



T , M , W , and H are four collinear points that exist such that $\overline{TW}$ and $\overline{MH}$ are congruent. $TW = 5y + 10$ and $MH = 9y - 46$. Four different students draw a diagram of the description. Each student's drawing is shown.

Name	Drawing
Miquel	
Harper	
Camila	
Liam	

- Determine the value of y .
- Use Miquel's drawing to determine the length of $\overline{WM}$ if $\frac{7}{3}WM = MH$.
- Use Harper's drawing to determine the length of $\overline{MW}$ if $\frac{3}{5}TM = MW$.
- Use Camila's drawing to determine the length of $\overline{MW}$ if $TM + MW + WH = 130$.
- Use Liam's drawing to determine the length of $\overline{WM}$ if $TH = 190\frac{1}{3}$.